

errors remain similar across different latent space dimensions (**Figure S1B**), the quality of the reconstructed embeddings varies substantially, as reflected in their performance on masked residue prediction (**Figure S1C**) and protein fitness prediction (**Figures S1D–E**).

We trained a standard SAE without the local similarity loss for comparison (see Methods). Both MotifAE and SAE achieve low reconstruction errors, with approximately 46 latent features activated per residue on average in both models (**Figure 2A–B**). To assess the quality of the reconstructed embeddings, we evaluated their performance in protein fitness prediction. We projected the reconstructed embeddings through the fixed ESM2 MLP layer to predict amino acid probability (**Figure 1**), calculated the wild-type marginal log-likelihood ratio (LLR; see Methods)^{22,24}, and used the LLR to estimate mutation effects. Across deep mutational scanning (DMS) experiments in ProteinGYM²³ and pathogenic/benign mutations in ClinVar²⁶ (see Methods), reconstructions from both MotifAE and SAE perform comparably to the original ESM2 embeddings (**Figure 2C–D**). These results demonstrate that both MotifAE and SAE preserve critical functional signals in ESM2 while requiring far fewer neurons to be activated in the latent space compared to the dimension of embedding.

The local similarity loss of MotifAE was introduced to encourage coherent latent feature activation. To evaluate this, we first analyzed feature activation similarity between nearby residues (measured using the absolute activation difference divided by the L1 norm of the activations: 0 for identical activations and ~2 for random sparse activations). MotifAE features were more similar among neighboring residues compared to SAE features (**Figure 2E**). We further examined the length distribution of activations and found that MotifAE features more frequently form clustered and contiguous activations than those of SAE, with a notably higher frequency of activations longer than four amino acids (**Figure 2F**). Together, these results showed that MotifAE preserves reconstruction quality and latent space sparsity, while the local similarity loss promotes locally coherent feature activations, with the potential to enable the identification of localized motifs and basic structural elements.

MotifAE captures known functional motifs.

To investigate how MotifAE latent features capture functional motifs, we compared their activations with experimentally validated motifs from the ELM² database, which covers six categories: ligand-binding, docking, post-translational modifications, targeting signals, degradation, and proteolytic cleavage sites. Each ELM motif has several experimentally verified functional regions, 270 motifs with at least 20 residues within verified regions were analyzed. We compared all MotifAE features against all ELM motifs, evaluating whether the feature activation scores could distinguish residues located inside versus outside the experimentally verified motif regions.

Considering the best-match feature for each motif, MotifAE demonstrates strong performance, achieving a median AUROC of 0.88 across the 270 motifs, significantly outperforming SAE (median AUROC 0.80) (**Figure 3A**). The best-match MotifAE features achieve AUROCs above 0.8 for 193 motifs and above 0.9 for 114 motifs out of 270 evaluated. One representative feature–motif pair is f13268 with the ELM motif MOD_TYR_CSK: the C-terminal phosphorylation motif in Src-family proteins targeted by the non-receptor tyrosine kinase Csk family²⁷. Across 12 proteins with known MOD_TYR_CSK motif in ELM, f13268 predicts residues in motif regions with an AUROC of 0.97. Notably, the highest activation of f13268 was observed for the phosphorylated tyrosine residue (**Figure 3B**), indicating that this feature captures the biological characteristics of the motif. Additionally, MotifAE performs consistently across structural contexts, with median AUROCs of 0.88 in ordered regions and 0.84 in disordered regions, compared to 0.79 and 0.76, respectively, for SAE (**Figure 3C**, see Methods).

Furthermore, we evaluated the specificity of the relationship between ELM motifs and MotifAE features. We defined a matched motif–feature pair as one with AUROC > 0.9, resulting in 322 matched pairs involving 114 ELM motifs and 146 MotifAE features (**Figure S2**). Among these, 47 ELM motifs are matched to a single MotifAE feature, whereas 67 motifs are matched to multiple features (**Figure 3D**). On the feature side, 91 MotifAE features match only one ELM motif (**Figure 3E**), whereas 55 features match multiple motifs. Among the MotifAE features that match multiple ELM motifs, f27416 is one of the most prominent: it matches 17 ELM motifs, all of which locate at the C-terminus of protein. Across all ELM motifs, f27416 achieves a median AUROC of 0.97 for 25 motifs exclusively located at the C-terminus but show no predictive power for other motifs (**Figure 3F**). Importantly, f27416 is not universally activated at the C-terminus of all proteins: among 320 proteins with known C-terminal motifs in ELM, 96% show f27416 activation, compared with only 39% of randomly sampled proteins (**Figure 3G**). Overall, these results indicate that MotifAE latent features align well with known functional motifs and perform substantially better than the standard SAE. MotifAE features also exhibit granularity: some capture motifs associated with very specific functions, while others represent more general or broadly shared motifs.

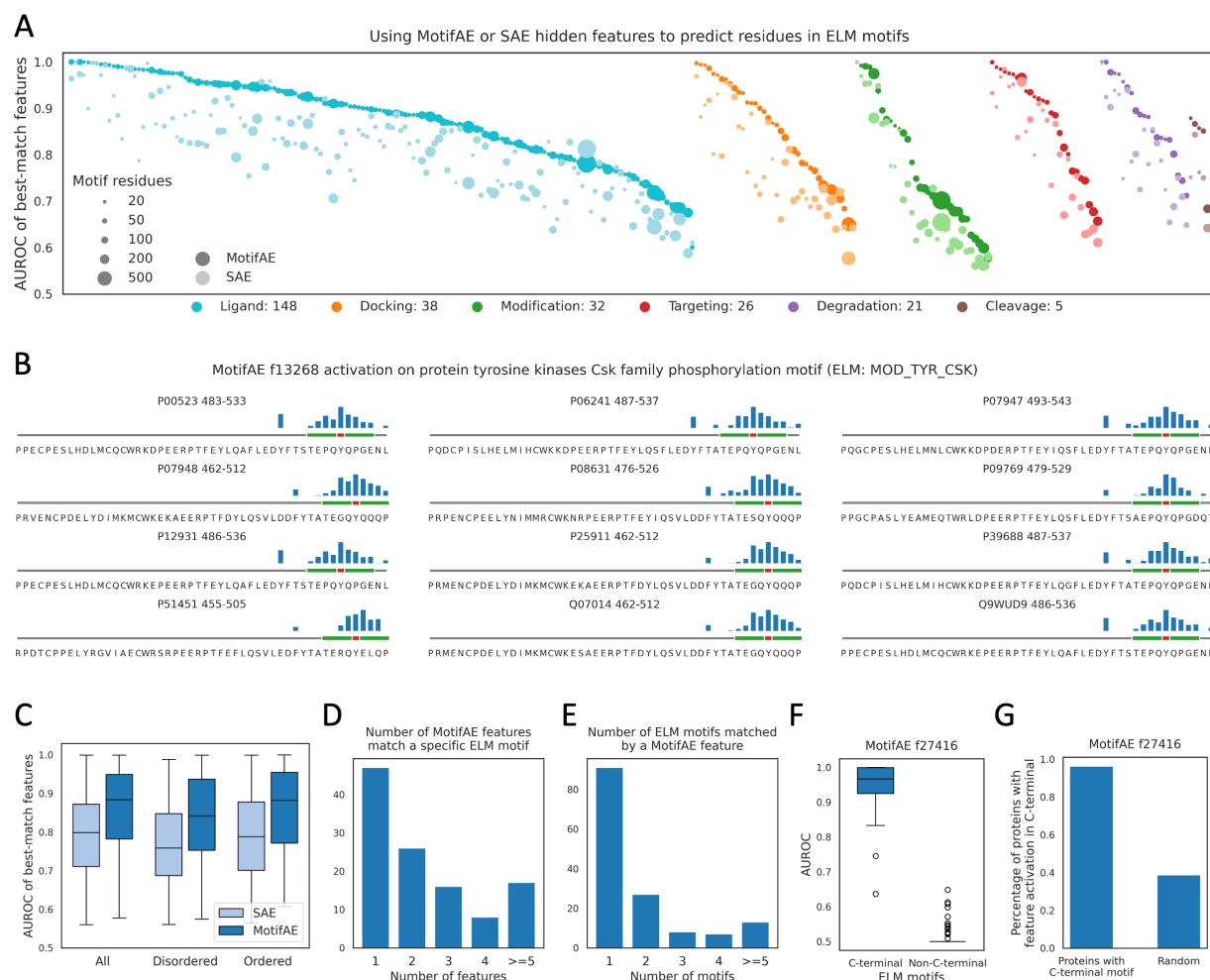


Figure 3. MotifAE captures known functional motifs.

(A) AUROC of the best-match feature for each ELM motif. Each column represents the performances of two models on an ELM motif; dots with dark colors indicate MotifAE, while dots with light colors indicate SAE. Dot size indicates the number of residues in motif regions, and colors represent different ELM categories. The six categories and the number of motifs in each are listed below the plot. (B) Feature activation along protein sequences. The height of the blue bars indicates the normalized feature activation values, the green region denotes the motif, and the red region marks the phosphorylated tyrosine residue. Only the C-terminal 50 residues of each protein are shown for clarity. (C) Distribution of AUROCs of best-match features for ELM motifs. Disordered and ordered regions were classified using IUPred3²⁸. The box extends from the first quartile to the third quartile, with a line marking the median. Whiskers span to the most extreme points within 1.5× the interquartile range. (D) Number of features match each ELM motif, (E) Number of motifs matched by each MotifAE feature, a motif-feature match is defined by AUROC > 0.9. (F) AUROC of MotifAE feature f27416 on C-terminal versus non-C-terminal ELM motifs. Non-C-terminal ELM motifs are defined as those with no verified regions within 10 residues of the C-terminus. (G) Percentage of proteins with activation of f27416 at the C-terminus. Proteins with known C-terminal motifs were obtained from the ELM dataset, random proteins were sampled from the evaluation set.

MotifAE captures homodimerization interfaces.

Next, we evaluated MotifAE on three-dimensional functional sites, focusing on homodimer interfaces. A previous study has shown that ESM2 captures evolutionary signals of homo-oligomer symmetry directly from single sequences²⁹. To construct the evaluation set, we retrieved homodimer structures from the PINDER³⁰ database and removed redundancy by clustering sequences at 30% identity (see Methods), yielding 1,565 non-redundant homodimers with at least five contact residues (contacts are defined by a heavy-atom distance cutoff of 5 Å). We used the activation value of each MotifAE feature to distinguish contact from non-contact residues within each protein. The best-match MotifAE feature for each homodimer achieves a median AUROC of 0.73 in identifying contact residues, outperforming SAE (0.69; **Figure 4A**). When stratifying homodimers by the number of contact residues (5–25, 25–50, >50), MotifAE performs particularly well on smaller homodimer interfaces, with a median AUROC of 0.82 for homodimers with 5–25

contact residues (SAE 0.76). Across homodimers with varying interface sizes, MotifAE consistently outperforms SAE. Representative homodimers with different number of contact residues are shown in **Figure 4B**, along with the activation values of their best-match MotifAE features.

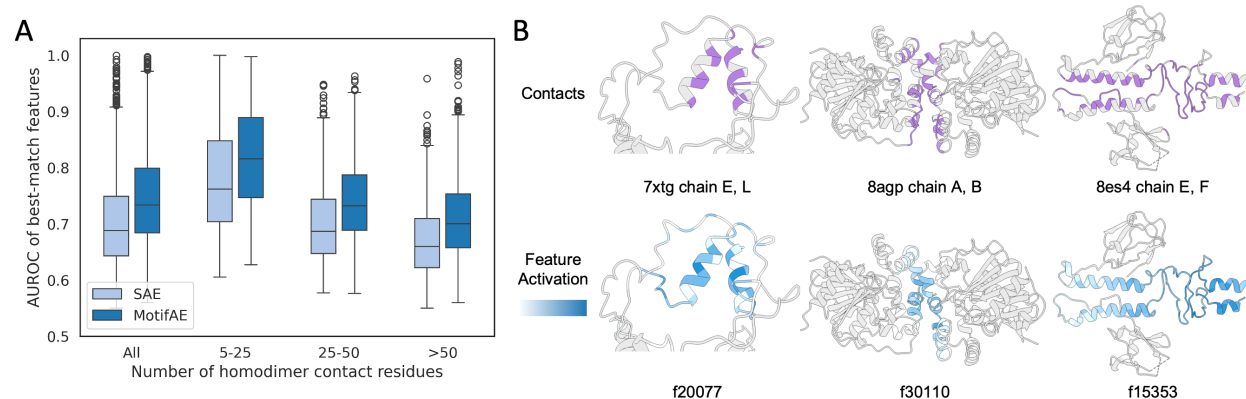


Figure 4. MotifAE captures homodimer interfaces.

(A) AUROC of the best-match MotifAE features for predicting each homodimer interfaces in the PINDER dataset. Homodimers are shown either collectively or grouped by the number of contact residues (5–25, 25–50, >50). (B) Representative homodimer structures. In the upper panel, contact residues are highlighted in purple. In the lower panel, the saturation of blue indicates the relative activation values of the best-match MotifAE feature.

The universe of sequence patterns of MotifAE features.

Next, we set out to characterize the sequence pattern of motifs captured by MotifAE features. Since functional motifs often exhibit distinct sequence preferences, we first analyzed the amino acid composition of activated peptides for each feature. Among the 40,960 features, 6,421 show coherent activations in the representative protein dataset²⁵. By recording activated peptides for each feature (see Methods), we found that MotifAE features display significantly biased amino acid preferences compared with random peptides (**Figure 5A**). Notably, approximately 400 features are almost exclusively activated by a single amino acid (**Figure 5B**; with one amino acid type accounting for >99% of all activations). To describe the sequence patterns, we constructed position-specific scoring matrices (PSSMs) for each MotifAE feature: activated peptides of each feature were aligned using GibbsCluster³¹, and the aligned cores were used to calculate the PSSMs (see Methods, PSSMs were set to a length of five residues for all features). The resulting PSSMs exhibited median KL divergence values (summed across the five residues) relative to the background amino acid distribution of 10.7 (**Figure 5C**). Among the PSSMs with well-defined sequence patterns, as indicated by higher KL divergence, some correspond to homopolymeric repeats (e.g., f13432 and f28775), while some exhibit multiple amino acid types (e.g., f2302, f36881, and f38747) (**Figure 5D**).

In the MotifAE decoder, each feature has a weight vector that is multiplied by its activation value to reconstruct the embedding. Thus, the decoder weights reflect the properties of features in the latent embedding space. Notably, features that have similar decoder weight vectors tend to have similar PSSMs (**Figure 5E**). For example, both f6643 and f20738 have an “H” residue centered between four “R” residues; both f2459 and f22016 capture an “LPLP” pattern; both f2182 and f23778 exhibit paired “Q” residues separated by a gap (**Figure 5D**). Together, by systematically characterizing the sequence patterns of MotifAE features, we found that they exhibit specific sequence preferences, we also found that these patterns are reflected in the decoder weight space.